This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	√	S1	√	S2
V810 S2EX	☐			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	9 Mar 2015	First Release	Chin Chee Yan
01	11 Mar 2015	Revise and add step	Chin Chee Yan
02	2 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG
03	28 Oct 2016	Content Update	ONG KEAN SHENG

Content

Filter Plate Assembly & Filter Height Sensor Adjustment

1. Unload board from machine if there is any. Go to X-ray source tab, turn off the X-ray. Switch off the key at Operator Console.

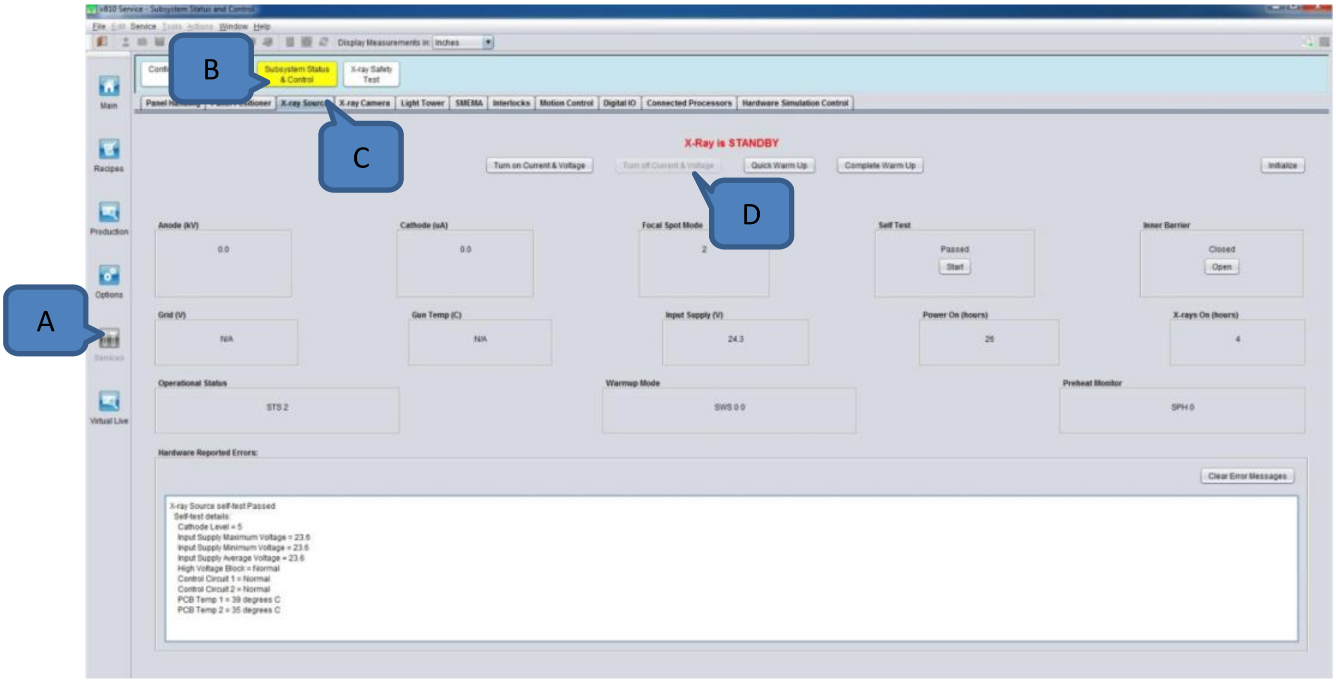


Figure 1

2. Go to Panel Handling tab, "Home" the AWA, trigger the inner barrier to "Open" position and change tube to "Down" position. Once complete home AWA, rail with will be 26 inch ++.

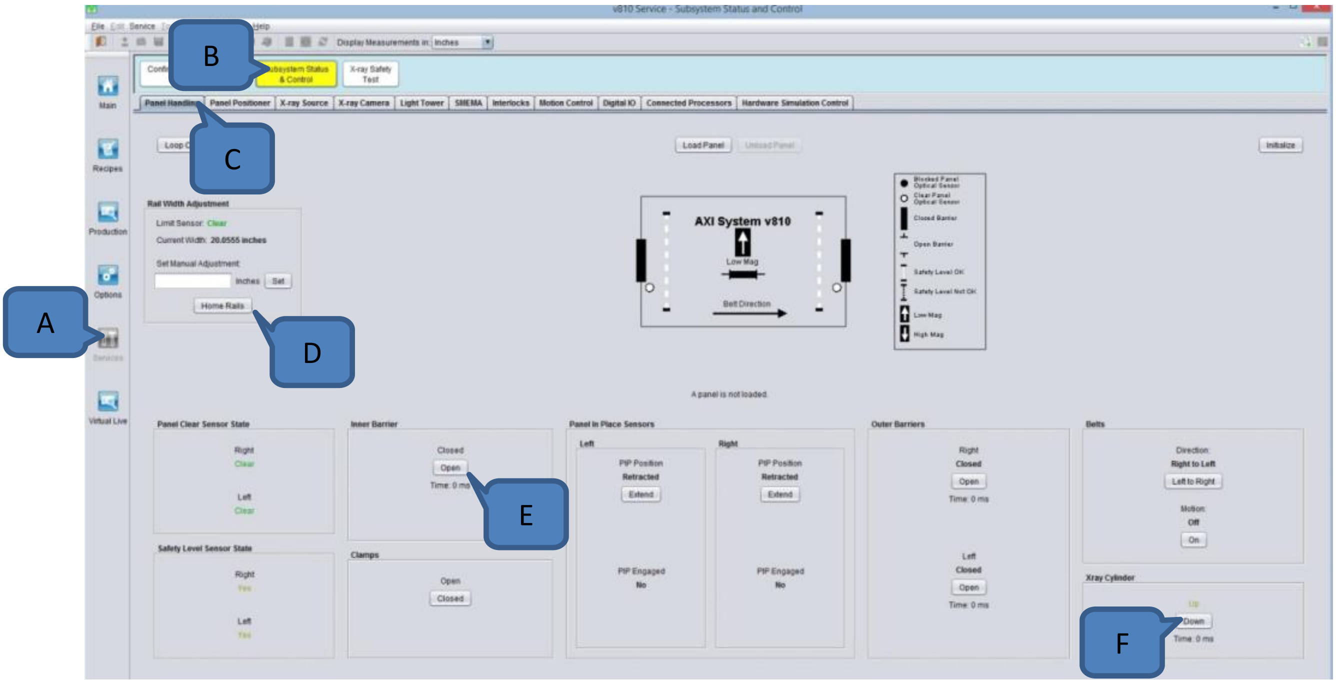


Figure 2

3. Open the front access doors, tube position should be at 13um position. Assembly the tube X-ray Cap (P/N, 4LAHW-051) at tube as shown with 2 Button Head Cap Screw M3x6 with spring washer and flat washer.

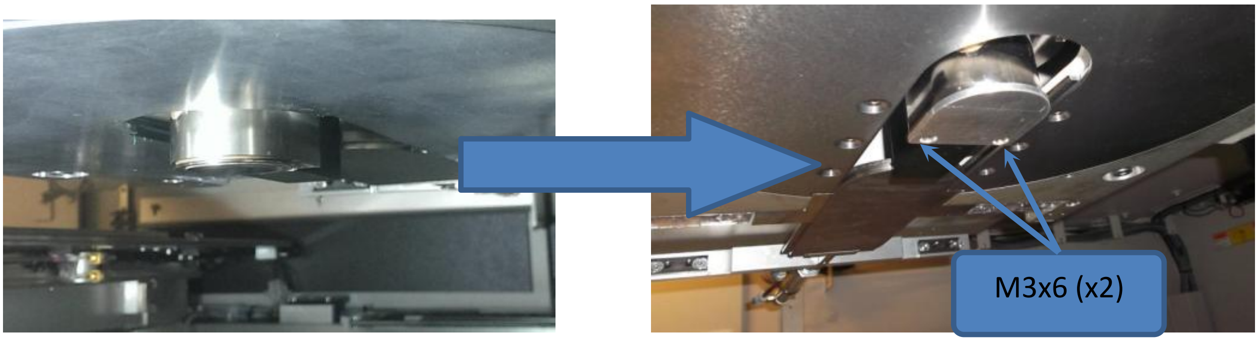


Figure 3

4. Return to panel Handling tab, change tube to "Up" position.

Figure 4

5. **Turn off the main air input** and manually exercise the inner barrier shutter plate. Ensure shutter plate is not contain friction with the tube target. Next, set the air pressure back to 85 PSI at main air regulator.

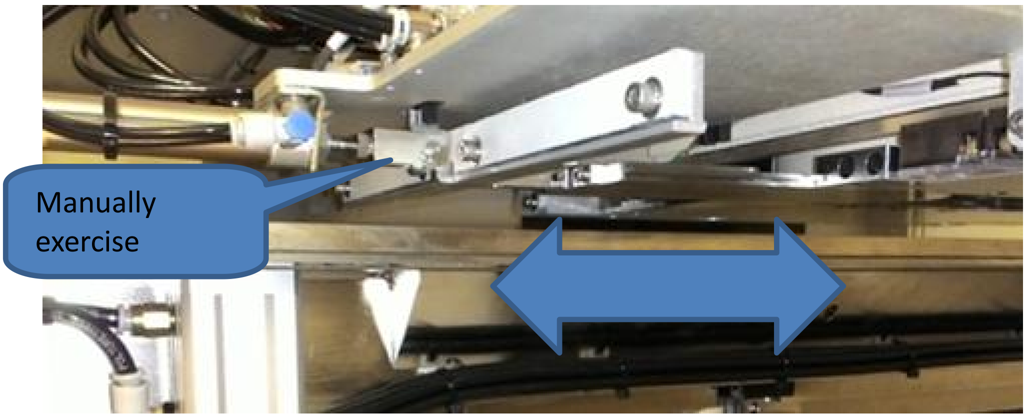


Figure 5

6. Prepare the HEIGHT JIG M13 FOR X-RAY FILTER (P/N, 4LASS-A306) as shown.

Figure 6

7. Place the jig at rail with to top clearance verification block face to tube , back to Panel handling tab, trigger the rail to Clamp position. Trigger tube to down position.

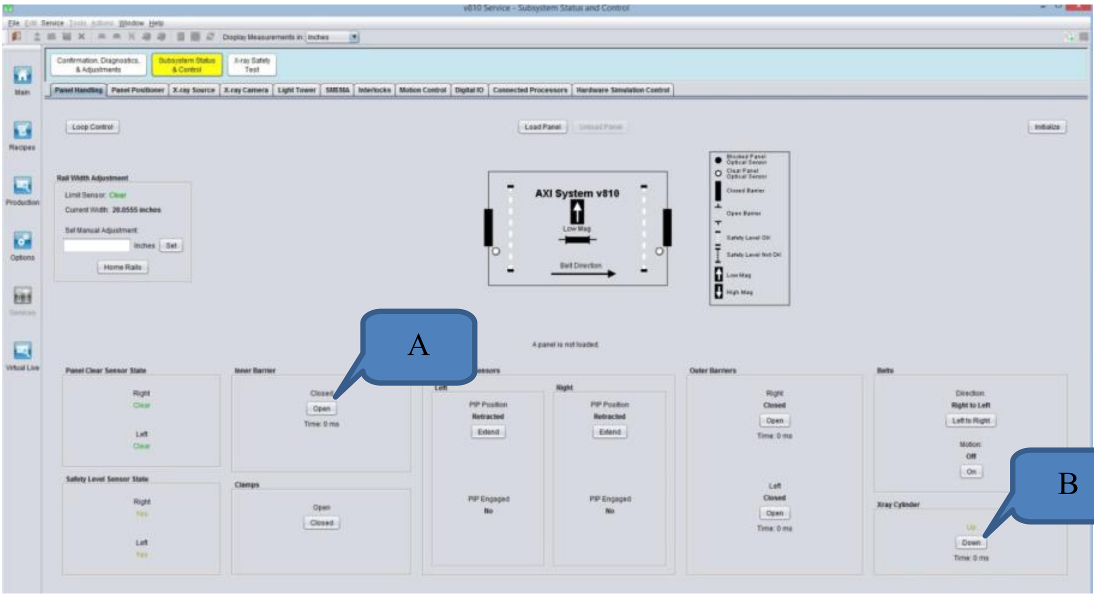


Figure 7

8. Manually move the stage across the tube. Ensure the top clearance block is not contact with tube.

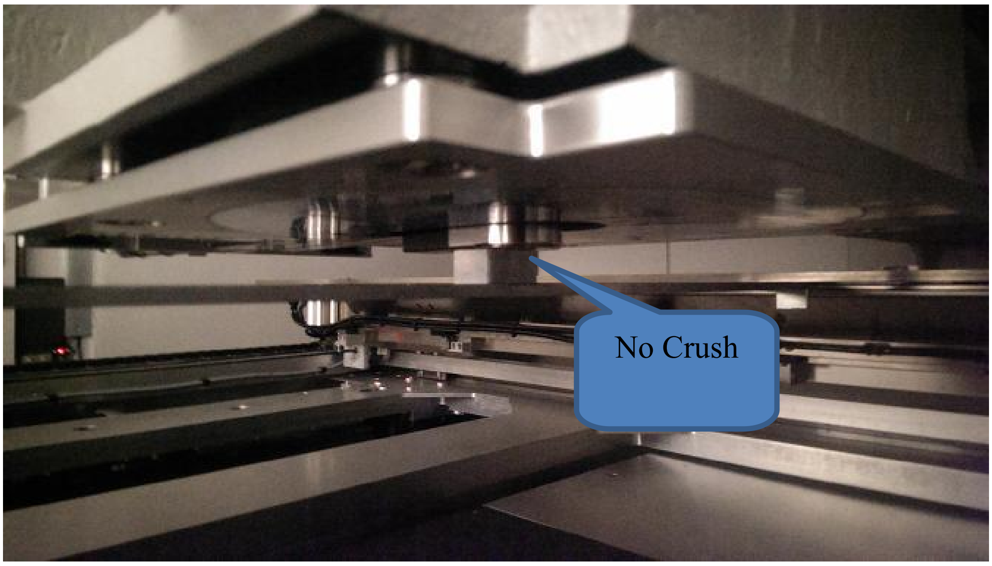


Figure 8

9. Return to Panel Handling tab, trigger tube to "Up" position and un-clamp the rail.

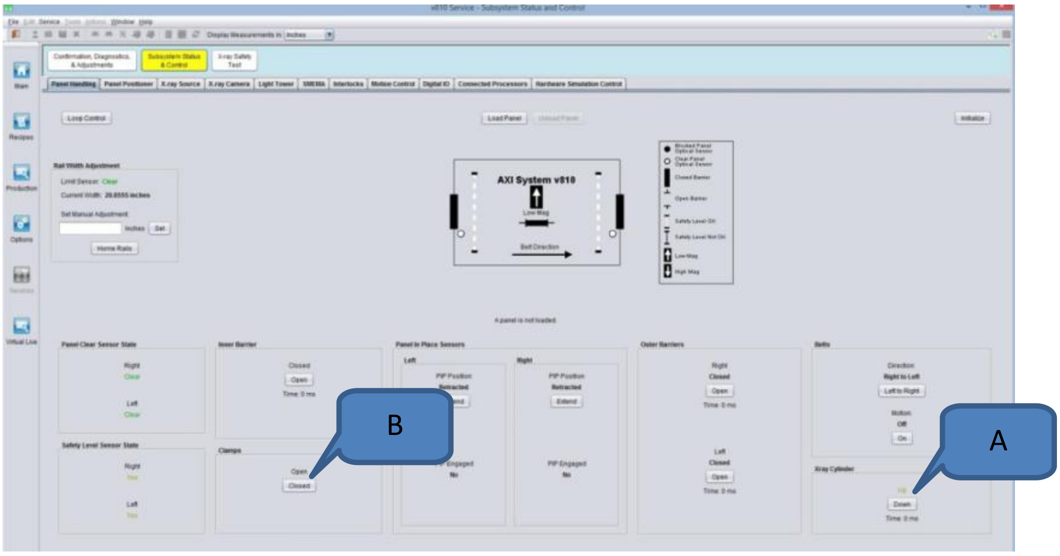


Figure 9

10. Flip the jig and place the jig near Left PIP. Once done, trigger the rail to "Clamp" position at Panel Handling tab.

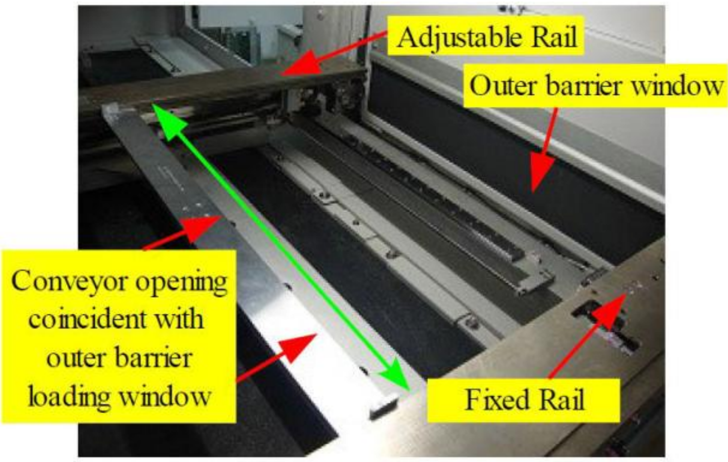


Figure 10

10. Manually move the stage until the Left Outer Barrier Side Filter Height Sensor Laser focus at Variable Mag Jig adjustment block as shown. (Remarks :Laser spot might focus at above new jig as previous height difference).

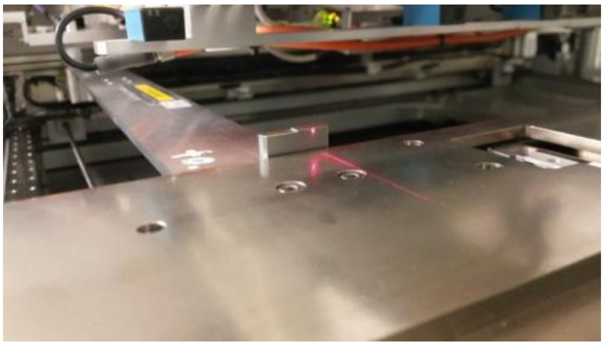


Figure 11

10. Remove SHCS M3×10 (x4) to remove Filter Sensor Module Cover for ease of tuning process.

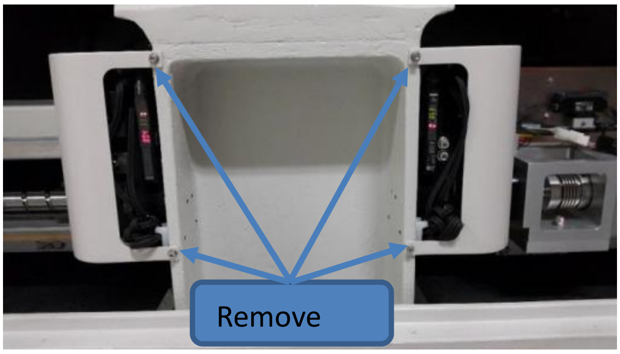


Figure 12

11. Use adjustable knob to adjust the angle and height of the filter sensor until the parallelism obtained as shown.

    11.1. Loosen Z-Height lock and position screw and adjust the height until the beam spot focus at Jig adjustment block. # May require to press the module downward.

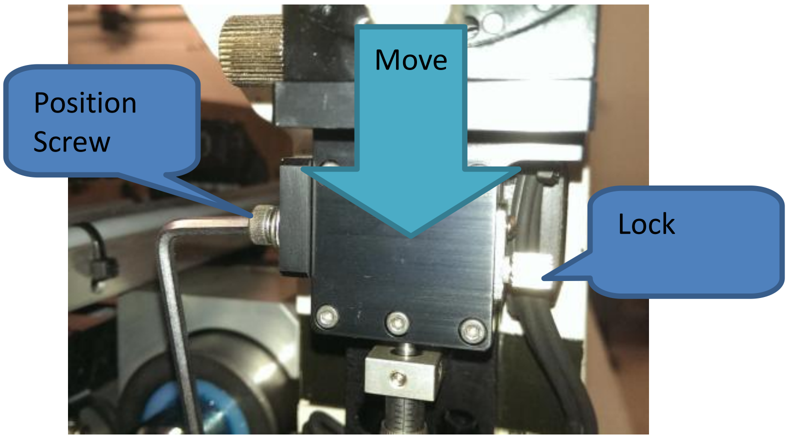


Figure 13

    11.2. Used the center point as reference (ease for precision fine tuning), ensure the spot beam focus at front and rear block center point (step between clear and block). Manually move X-Axis ball screw coupling and check laser spot is approximately same level within front and rear block as step 12.

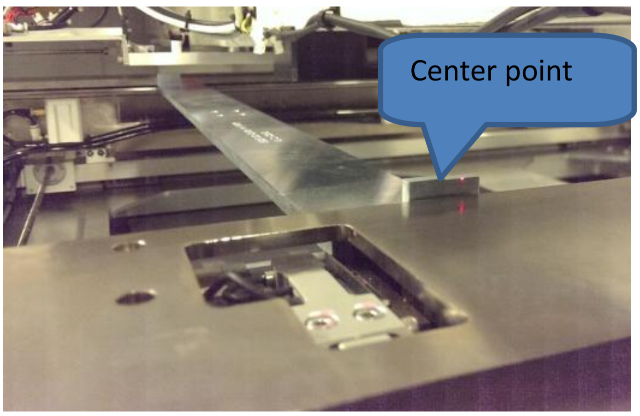


Figure 14

11.3. Once complete, tighten the lock. (# Suggest tighten the lock 1st when the height approximately reach to same level, minor turn the knob again to avoid threshold fluctuate issue.)

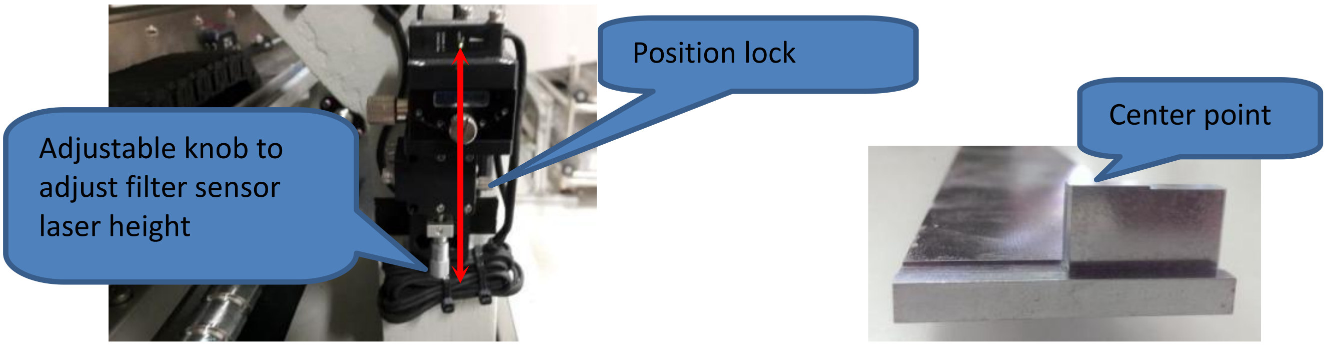


Figure 15

12. The sensor laser position for the front and rear adjustment block (Use the centre point between 13 and 13.5 mm as 1st reference) must focus at same level.

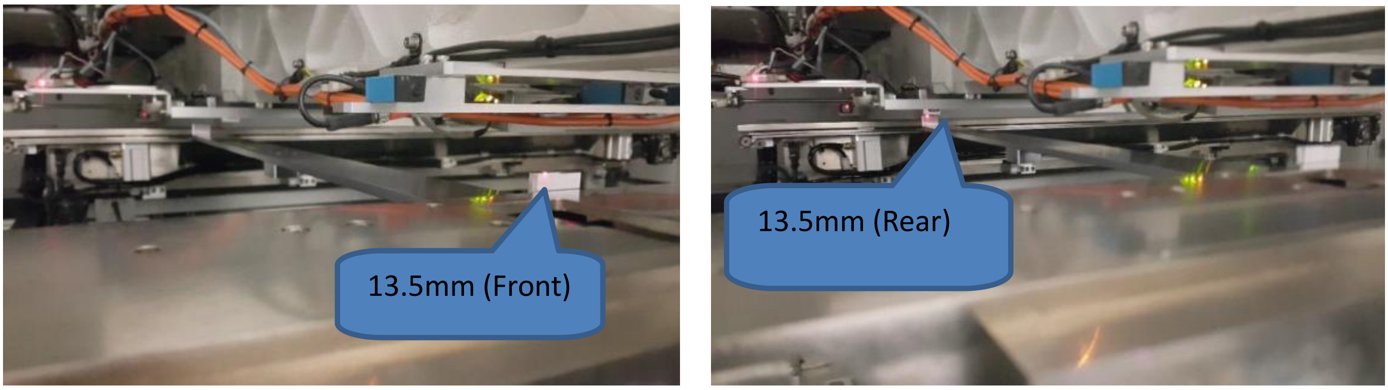


Figure 16

13. Check the parallelism efficiency by checking the value (Red) of the sensor amplifier for precision fine tuning. The range for the front and rear variable mag jig should be almost the same as shown in the pic. Remarks: Difference in value for the front and back variable mag jig **shouldn't exceed more than 50.**

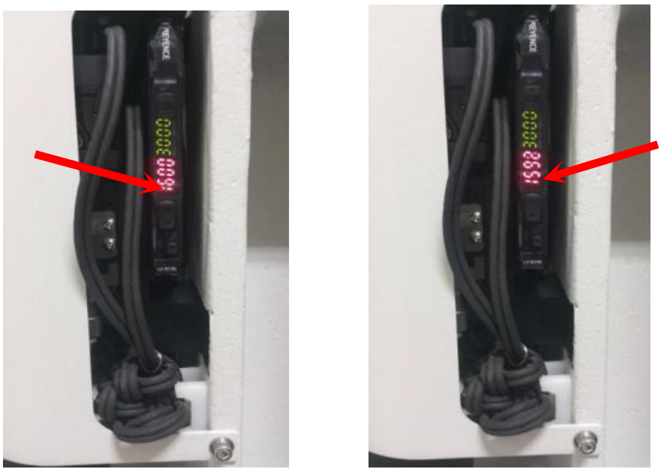


Figure 17

Front Rear

14. If variance of intensity of Amplifier exceed 50, slightly adjust the beam angle. (# Suggest tighten the lock 1st when the height approximately same level, minor turn the knob to avoid threshold fluctuate issue.)

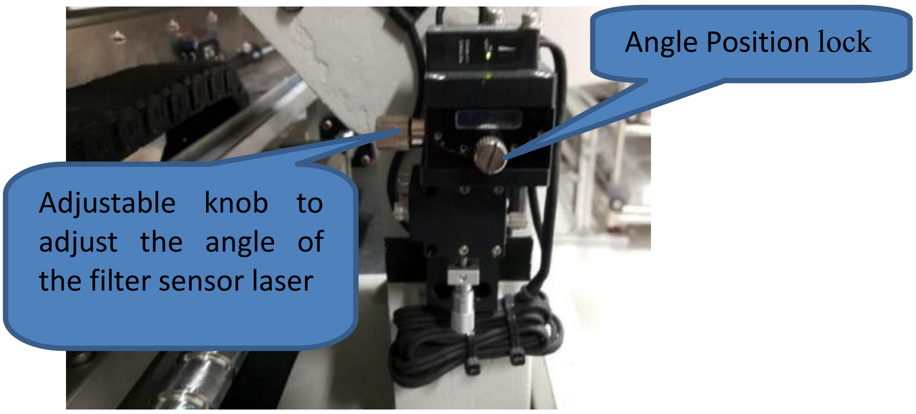


Figure 18

15. Adjust the height of the sensor until meet the condition shown in the Table 1.

Orientation	Jig Height	Amplifier Value
A (Fix Rail)	13.0	>3000
	13.5	<2400
B (Adj Rail)	13.0	>3000
	13.5	<2400

Table 1

16. Filter Sensor Spot Beam straightness explanation sample as below,

Figure 19

- If **Fix Rail`s block intensity is lower** than Adjustable Rail`s block intensity, mean the laser spot beam is pointing upward. Require to adjust the **FHSM to upper position and lower the beam angle**.
- If the **Fix Rail`s block intensity is higher** than Adjustable Rail`s block intensity, mean the laser spot beam is pointing downwards. Require to adjust he **FHSM to lower position and upper the beam angle**.

- Repeat step 11~ 14 until intensity is within spec as Table 1.
- Variation Reading for the Fix Rail and Adjustable Rail 13.5mm and 13 mm should not exceed 50

17. Once adjustment complete, counter check the hardware situation and ensure system meet condition as table 2.

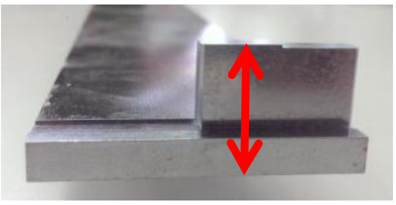
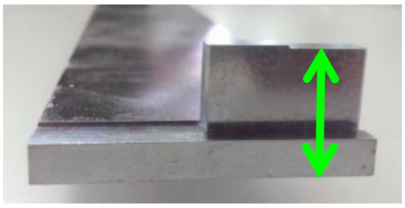
Jig Height Image	Height	Filter Sensor Status	DIO Input Bit	Term
	**13.5**			➢ Only remain 1 LED at sensor is emitted. ➢ DIO input bit show **"Block".** ➢ Amplifier Value (Red) **less than 3000**
	**13**			➢ 4 LED at sensor are emitted. ➢ DIO input bit shown **"Clear".** ➢ Amplifier Value (Red) **more than 3000**

Table 2

18. Once Left FHSM adjustment complete, return to Panel Handling tab, un-clamp the rail and re-position the jig near to right PIP and clamp again.
19. Slowly push the stage until the Right Filter Height Laser hit the Jig. Repeat from Step 10 to Step 17 for the Right Filter Height Sensor Module.

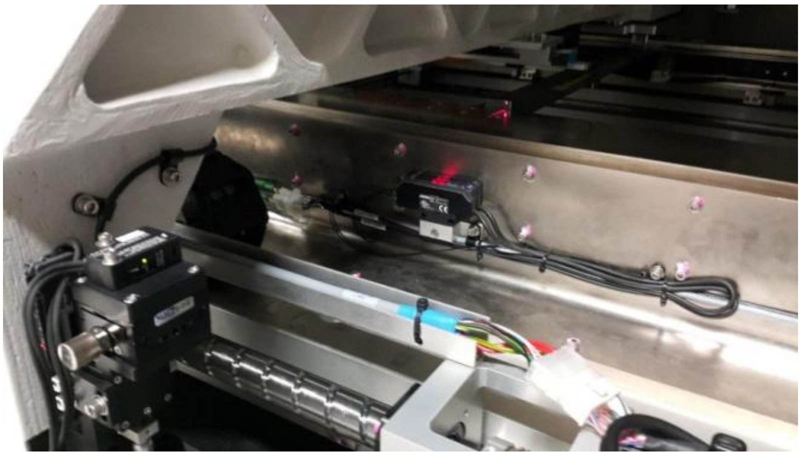


Figure 20

20. Once left and right FHSM adjustment complete, assemble the Filter Sensor Module Cover into origin position and verify again the result by refer to Table 2. Remove the jig as well.

Filter Sensor Amplifier Configuration

21. Open Amplifier top cover, the button and display panel will show as attach. Ensure channel 1(Output 1) is set.

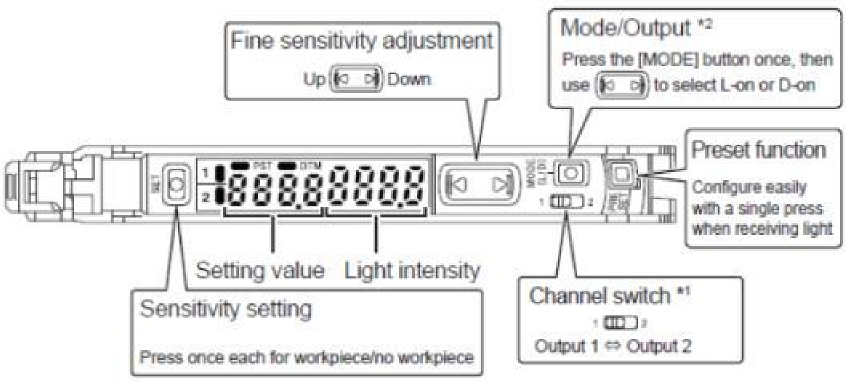


Figure 21

22. Press the “Mode” and hold button for more than 3 seconds to enter the setting mode.

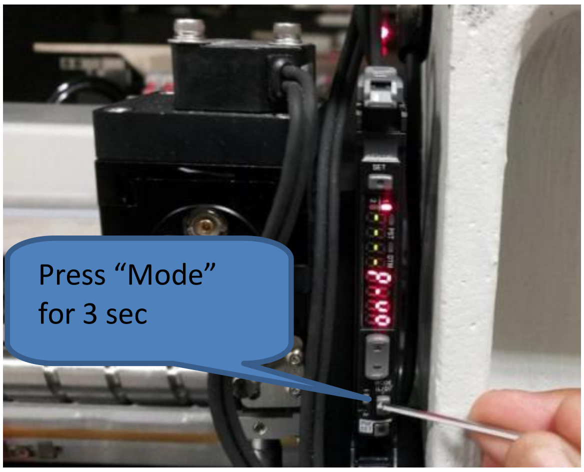


Figure 22

23. Press “Up” or “Down” until the “SuPr” mode is selected as shown in the Figure 23.

Remarks : Maximum intensity can reach more than 1100 when the laser is directly shooting at the reflector. If the reading is less than 1100, clean the reflector surface as shown in Figure 24

Figure 23

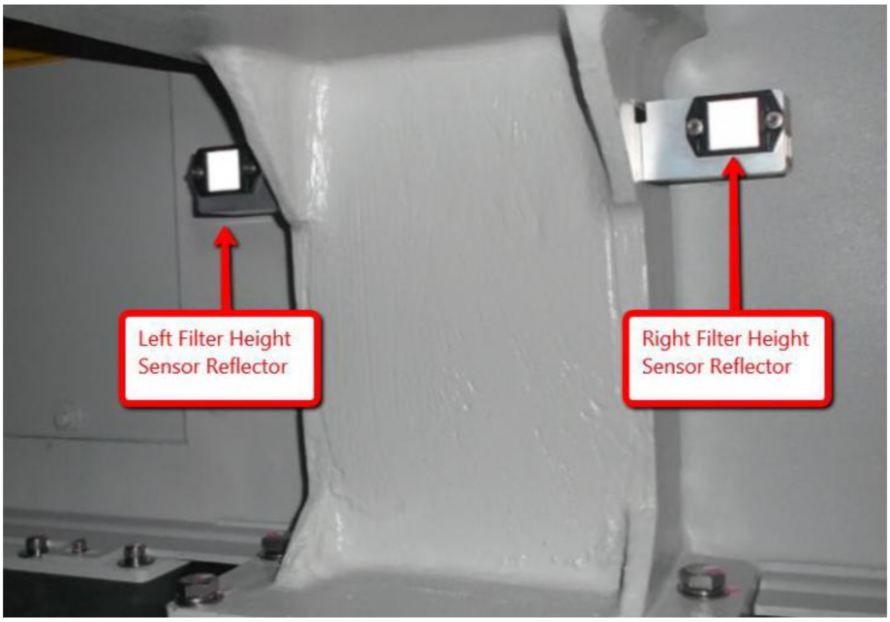


Figure 24

24. Press the “mode” button followed by “Up” or “Down” until the “Std” option is selected as shown in the Figure 25

Figure 25

25. Press the “mode” button followed by “Up” or “Down” button to select the “Func” as shown in the Figure 26

Figure 26

26. Press the “mode” button followed by “Up” or “Down” and select the “on-d” option as shown in the Figure 27

Figure 27

27. Press the “mode” button followed by “Up” or “Down” button until value reach 5000 as shown in the Figure 28

Figure 28

28. Press the “mode” button followed by “Up” or “Down” button to select the “dEtc” mode as shown in the Figure 29. Press “mode” button again followed by “Up” or “Down” button to select the “Std” option as shown in the Figure 29.

Figure 29

29. Press the "mode" button followed by "Up" or "Down" to select the "In oFF" mode as shown in the Figure 30.

Figure 30

30. Press "mode" button again to exit the setting
31. Press "L/D" button followed by "Up" or "Down" button to select the "L-on" mode as shown in the Figure 31

Figure 31

32. After few seconds, the amplifier will return to the main menu

    Set the threshold for the sensor amplifier to 450 using "Up" or "Down" button as shown in the Figure 32

Figure 32